Practical Interview Test (2 Hours)

Task 1: Models

1. Create a User model with fields: id, name, email, password [You can use the default user model in Laravel].
2. Create a Product model with fields: id, name, description, price.
3. Create a UserRating model with fields: id, user_id (foreign key), product_id (foreign key), rating, rating_datetime.

Task 2: Migrations

1. Create migrations for the users, products, and user_ratings tables.
2. Define relationships between User, Product, and UserRating models.
3. Add dummy data to the users and products tables.

Task 3: Controllers

- Create a controller named RatingController. o Implement methods for rating a product, removing a rating, and changing a rating.
- Ensure that a user can only create a product rating once, any change to the rating will update the current rating. (i.e. in the user_ratings table, the combination of user_id and product_id is unique)
- Create method to display a list of products, each product should return a ratings column, which is the average of all the ratings of that product.
    1. Also, another column user_rating which displays the product rating of the user requesting.
    2. Also add another field called time_passed, which will return the difference in minutes between the displayed time and rating_datetime.
    3. Add another field called active_time, which if time_passed is greater than 30minutes, it returns active, else inactive.

Task 4: API Routes

- Define API routes for rating a product, removing a rating, and changing a rating and listing a.

Task 5: Validation

1. **Validation Rules:**
    - Implement validation rules to ensure that the rating value is within a valid range of 1 to 5.

Task 6: Push to Git

1. **Create a public git repository on github:**

   - push your code to the public repository o
     Share your repository link with Mr.Thomas

(BONUS TASK):

--Create an endpoint to register a new patient with Gpitg Hospital. send the registration request to the provided external endpoint. Once a response is returned, display Check_In_Date_And_Time returned with message successfully

end point : http://41.188.172.204:3033/patient-registration


```
{
"Sponsor_ID": "1",
"Patient_Name": "ngenzi ngenzi",
"Date_Of_Birth": "2022-07-02",
"Gender": "Male",
"Visit_Type_ID": "1",
"Type_Of_Check_In": "1",
"branchId": "1",
"Employee_ID": "46",
"pf3": null,
"Diceased": "no",
"Referral_Status": null
}
```


Authentication

- Create a login endpoint to login the user. It should return a token
- Authorize each endpoint except the login endpoint